

Echelon Forms

Geometric Algorithms Lecture 2

Practice Problem

None today...

3 planes in $\mathbb{R}^3$, none parallel, no solution

Outline

- » Introduce echelon forms as matrices which "look like" solutions
- » Learn to "read off" a solution from an echelon form

Keywords

leading entries

echelon form

(row-)reduced echelon form (RREF)

pivot positions

pivot columns

free variables

basic variables

general form solutions

forward elimination

back substitution

Recap

Recall: Linear Systems (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Recall: Linear Systems (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Does a system have a solution?

How many solutions are there?

What are its solutions?

Recall: Matrix Representations

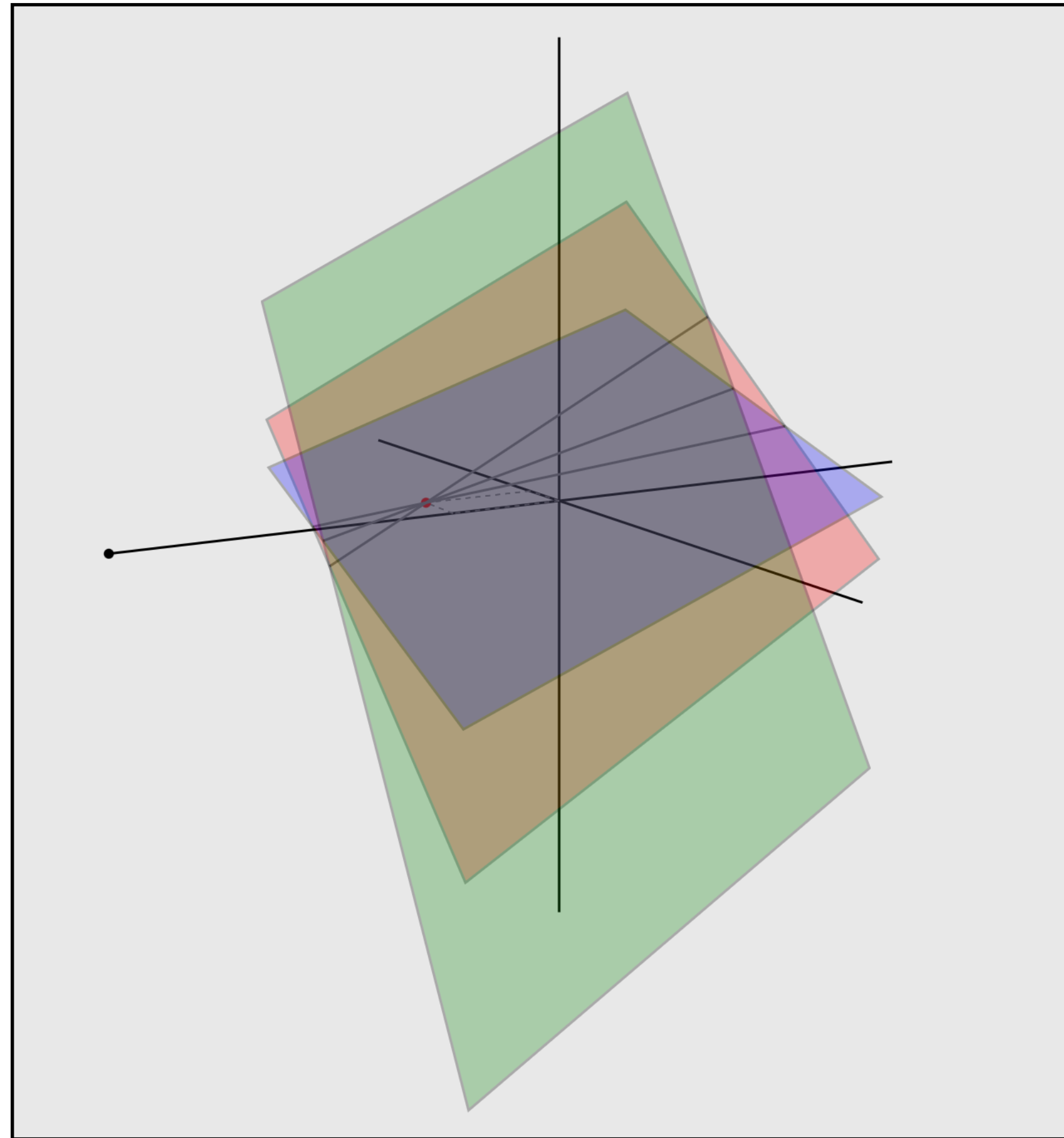
$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Recall: Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

Recall: Linear Systems, Geometrically



Recall: Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

Recall: Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

These are the **only** options

where we left off...

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Another Approach

Elimination

Eliminate x from the EQ2 and solve for y

Eliminate y from EQ1 and solve for x

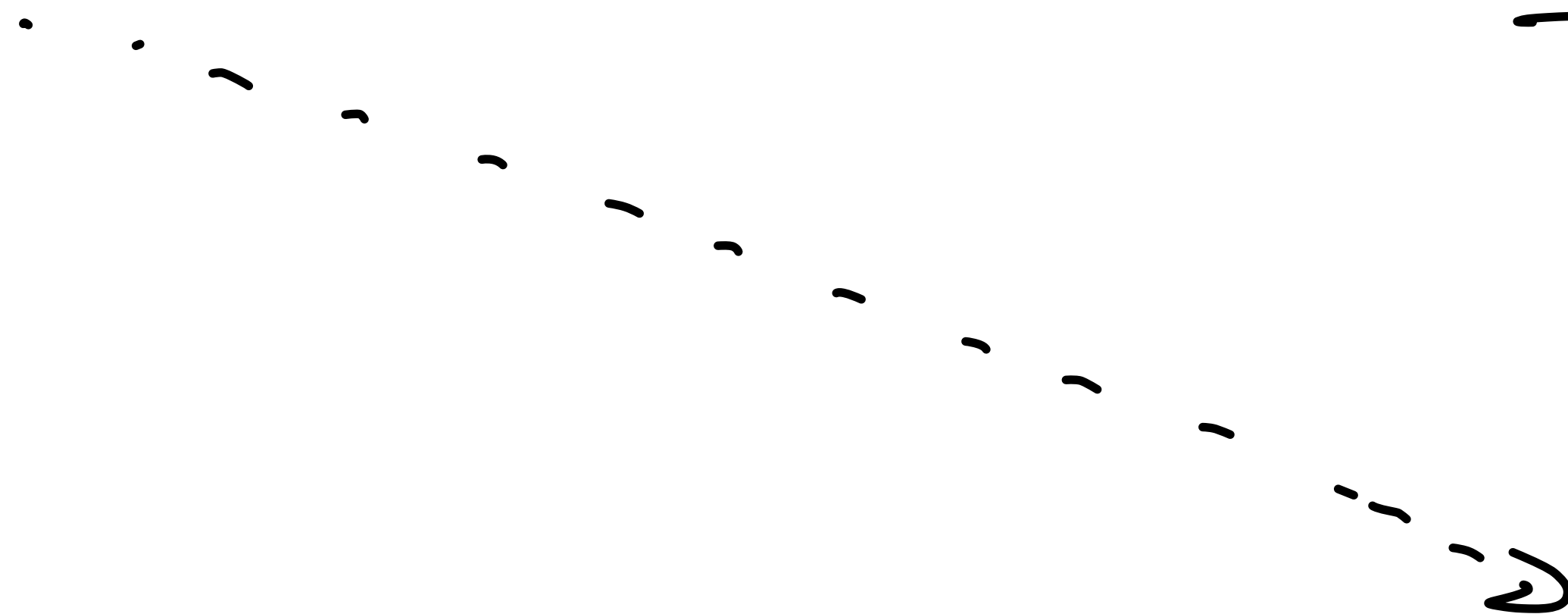
Back-Substitution

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$6x - 2y = 4$$

$$\begin{array}{r} 2x + 3y = -6 \\ + \quad 4x - 5y = 10 \\ \hline 6x - 2y = 4 \end{array}$$



$$x = ?$$

$$y = ?$$



still linear
system

Solving Systems with Two Variables

$$\begin{array}{l}
 \text{EQ}_1: 2x + 3y = -6 \\
 \text{EQ}_2: 4x - 5y = 10 \\
 \quad \quad \quad \begin{array}{ccc} -2x & -6y & +12 \end{array} \\
 \Downarrow \\
 \begin{array}{r} 2x + 3y = -6 \\ -11y = 22 \end{array} \\
 \Downarrow \\
 \begin{array}{r} 2x + 3y = -6 \\ y = -2 \end{array}
 \end{array}$$

$$\text{EQ}_2 - 2\text{EQ}_1$$

$$\text{EQ}_2 / -11$$

$$\text{EQ}_1 - 3\text{EQ}_2$$

formed
elim.

$$2x + 3y = -6$$

$$4x - 5y = 10$$

$$2x = 0$$

$$y = -2$$

$$\text{EQ}_1 / 2$$

$$x = 0$$

$$y = -2$$

still linear
system

Solving Systems as Matrices

How does this look with matrices?

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the same solutions

Solving Systems as Matrices

How does this look with matrices?

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the same solutions

Can we represent these intermediate steps as operations on matrices?

Let's look back at this...

$$\begin{array}{l} R_1 \\ R_2 \end{array} \begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \\ -4 & -6 & 12 \end{bmatrix} \quad R_2 \leftarrow R_2 - 2R_1 \quad \text{row operation}$$

$$2x + 3y = -6$$

$$4x - 5y = 10$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \\ -4 & -6 & 12 \end{bmatrix} \quad \begin{array}{l} 2x + 3y = -6 \\ -11y = 22 \end{array}$$

$$R_2 \leftarrow R_2 / -11$$

$$\begin{bmatrix} 2 & -3 & -6 \\ 0 & 1 & -2 \\ -4 & -6 & 12 \end{bmatrix} \quad R_1 \leftarrow R_1 - 3R_2$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix} \quad R_1 \leftarrow R_1 / 2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix} \quad \begin{array}{l} x = 0 \\ y = -2 \end{array}$$

Elementary Row Operations

scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

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scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

These operations don't change the solutions

Scaling

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \leftarrow 4R_2} \begin{bmatrix} 1 & 2 & 3 \\ 16 & 20 & 24 \\ 7 & 8 & 9 \end{bmatrix}$$

Scaling

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_1 \leftarrow 2R_1$$


$$\begin{array}{l} 4x + 6y = -12 \\ 4x - 5y = 10 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 6 & -12 \\ 4 & -5 & 10 \end{bmatrix}$$

Replacement

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 15 & 18 & 21 \end{bmatrix}$$

Handwritten annotations in red:

- $+8$ next to 7
- $+10$ next to 8
- $+12$ next to 9

Replacement

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_2 \leftarrow R_2 + R_1$$



$$\begin{array}{l} 2x + 3y = -6 \\ 6x - 2y = 4 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 6 & -2 & 4 \end{bmatrix}$$

Interchange

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

Interchange

$$\begin{aligned}2x + 3y &= -6 \\4x - 5y &= 10\end{aligned}$$

$$R_1 \leftrightarrow R_2$$



$$\begin{aligned}4x - 5y &= 10 \\2x + 3y &= -6\end{aligned}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -5 & 10 \\ 2 & 3 & -6 \end{bmatrix}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

Example: Row Reductions

$$\begin{array}{ccc} & R_2 \leftarrow R_2 - 2R_1 & \\ \begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} & \longrightarrow & \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix} \\ & R_2 \leftarrow R_2 / (-11) & \\ & \longrightarrow & \begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix} \end{array}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / (-11)$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

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$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

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$$R_1 \leftarrow R_1 / 2$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

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$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

elimination

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

back-substitution

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Row Equivalence

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Row Equivalence

$$A \sim B$$

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

We can compute solutions by sequence of row operations

IMPORTANT

Row equivalent augmented
matrices represent linear system
with the same solution set

Example

Exercise.

$$x - 2y + z = 5$$

$$2y - 8z = -4$$

$$6x + 5y + 9z = -4$$

Q: When are we done reducing?
What's the "target" matrix?

A: when we are able to
"read off" a solution

Motivating Questions

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What matrices "represent solutions"? (solutions are easy to "read off"?)

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echelon forms

What matrices "represent solutions"? (solutions are easy to "read off"?)

How does the number of solutions affect the shape of these matrices?

How do we use row operations to get to those matrices?

Unique Solution Case

Unique Solution Case

$$\begin{bmatrix} 2 & -3 & 5 & 11 \\ 2 & -1 & 13 & 39 \\ 1 & -1 & 5 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$x = 1$$

$$y = 2$$

$$z = 3$$

Unique Solution Case

$$\begin{bmatrix} 2 & -3 & 5 & 11 \\ 2 & -1 & 13 & 39 \\ 1 & -1 & 5 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$x = 1$$

$$y = 2$$

$$z = 3$$

The Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1s along the diagonal

0s elsewhere

Unique Solution Case

coefficient matrix

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

a system of linear equations whose **coefficient matrix** is the identity matrix represents a unique solution

No Solution Case

No Solution Case

$x + y + z = 1$
 $x + y + z = 2$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

No Solution Case

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

two parallel
planes

No Solution Case

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

two parallel planes

row representing $0 = 1$
 $0x + 0y + 0z = 1$

No Solution Case

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

row representing $0 = 1$

a system with no solutions can be reduced to a matrix with the row

$$0 \ 0 \ \dots \ 0 \ 1$$

Infinite Solution Case

Infinite Solution Case

$$\begin{bmatrix} 2 & 4 & 2 & 14 \\ 1 & 7 & 1 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x + z = 2$$

$$y = 1$$

$$(0, 1, 2) \quad (\pi, 1, 2 - \pi)$$

$$(3, 1, -1)$$

$\vdots$

$\vdots$

Infinite Solution Case

$$\begin{bmatrix} 2 & 4 & 2 & 14 \\ 1 & 7 & 1 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

Infinite Solution Case

$$\begin{bmatrix} 2 & 4 & 2 & 14 \\ 1 & 7 & 1 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

a system with infinity solutions can be reduced to a system which leaves a variable unrestricted

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\ x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\x_2 &= 1\end{aligned}$$

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$$x_1 = 2$$

$$x_2 = 1$$

$$x_3 = 0$$

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$x_1 = 1.5$$

$$x_2 = 1$$

$$x_3 = 0.5$$

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\ x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$x_1 = 20$$

$$x_2 = 1$$

$$x_3 = -18$$

Infinite Solution Case

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

Infinite Solution Case

$$\begin{aligned}x_1 + x_3 &= 2 \\ x_2 &= 1\end{aligned}$$

it doesn't matter
what x_3 is if we
want to satisfy
this system of
equations

$$\begin{aligned}x_1 &= 2 - x_3 \\ x_2 &= 1 \\ x_3 &\text{ is free}\end{aligned}$$

general form

In Sum

- none** reduces to a system with the equation $0 = 1$
- one** reduces to a system whose coefficient matrix is the identity matrix
- infinity** reduces to a system which leaves a variable unrestricted

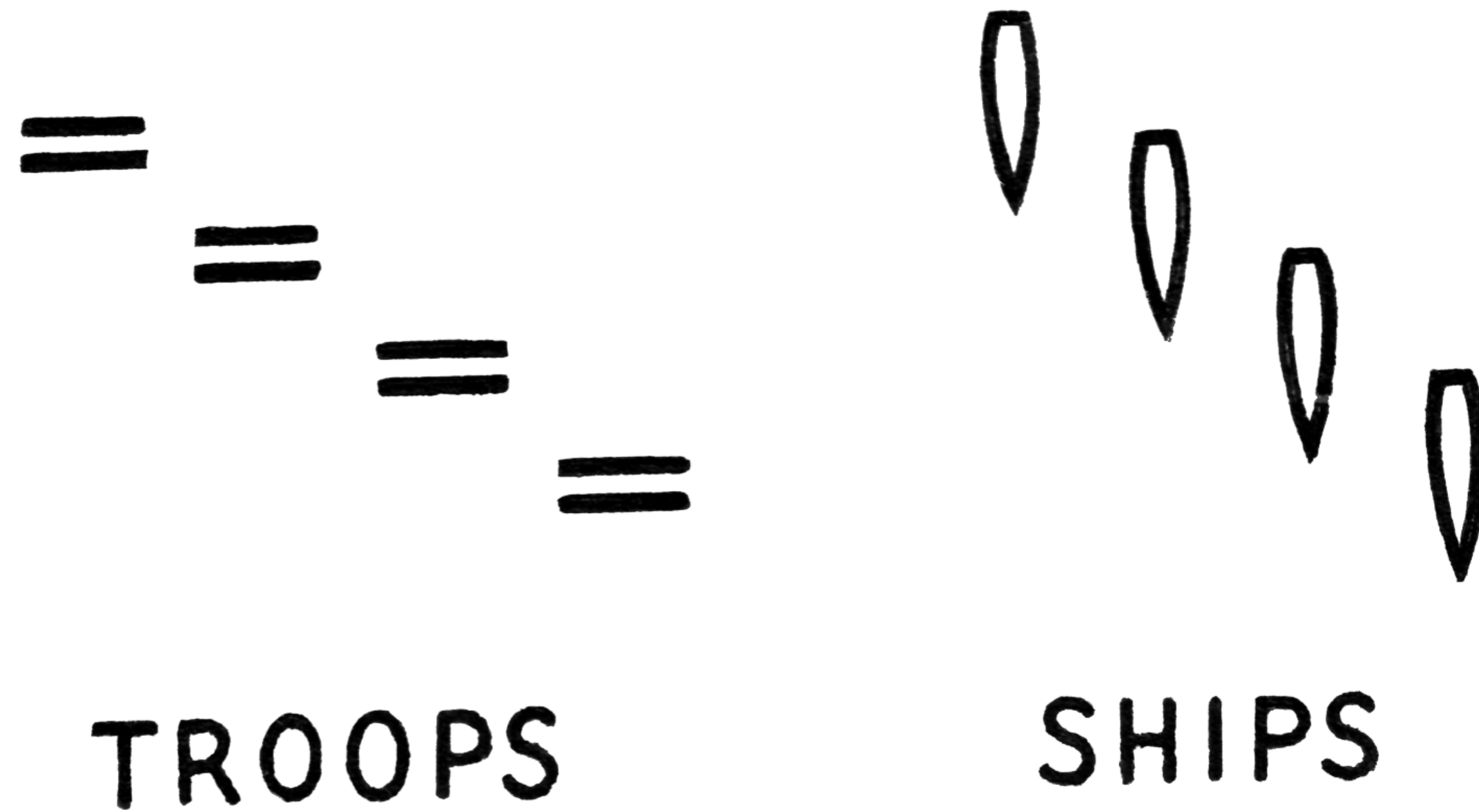
In Sum

- `none` reduces to a system with the equation $0 = 1$
- `one` reduces to a system whose coefficient matrix is the identity matrix
- `infinity` reduces to a system which leaves a variable unrestricted

Ideally, we want one *form* that handles all three cases

Echelon Form

The Picture (and a bit of history)



Echelon Form (Pictorially)

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\blacksquare$ = nonzero, $*$ = anything

Leading Entries

Df. the *leading entry* of a row is the first nonzero value

$$\begin{bmatrix} \boxed{1} & 2 & 0 \\ 0 & 0 & \boxed{1} \\ 0 & \boxed{1} & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

no leading entries

Echelon Form

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Df. A matrix is in *echelon form* if

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1. The leading entry of each row appears to the right of the leading entry above it

Echelon Form

Df. A matrix is in *echelon form* if

1. The leading entry of each row appears to the right of the leading entry above it
2. Every all-zeros row appears below any non-zero rows

Echelon Form (Pictorially)

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\blacksquare$ = nonzero, $*$ = anything

Echelon Form (Pictorially)

next leading entry
to the right

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

all-zero rows at
the bottom

$\blacksquare$ = nonzero, $*$ = anything

Question

Is the identity matrix in echelon form?

A hand-drawn diagram illustrating the identity matrix in echelon form. The matrix is enclosed in large square brackets and contains three rows. Each row has a leading 1 in a purple box, followed by zeros. Purple arrows point from the zero in the first row to the 1 in the second row, and from the zero in the second row to the 1 in the third row, showing the staircase pattern of the leading ones.

$$\begin{bmatrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1} \end{bmatrix}$$

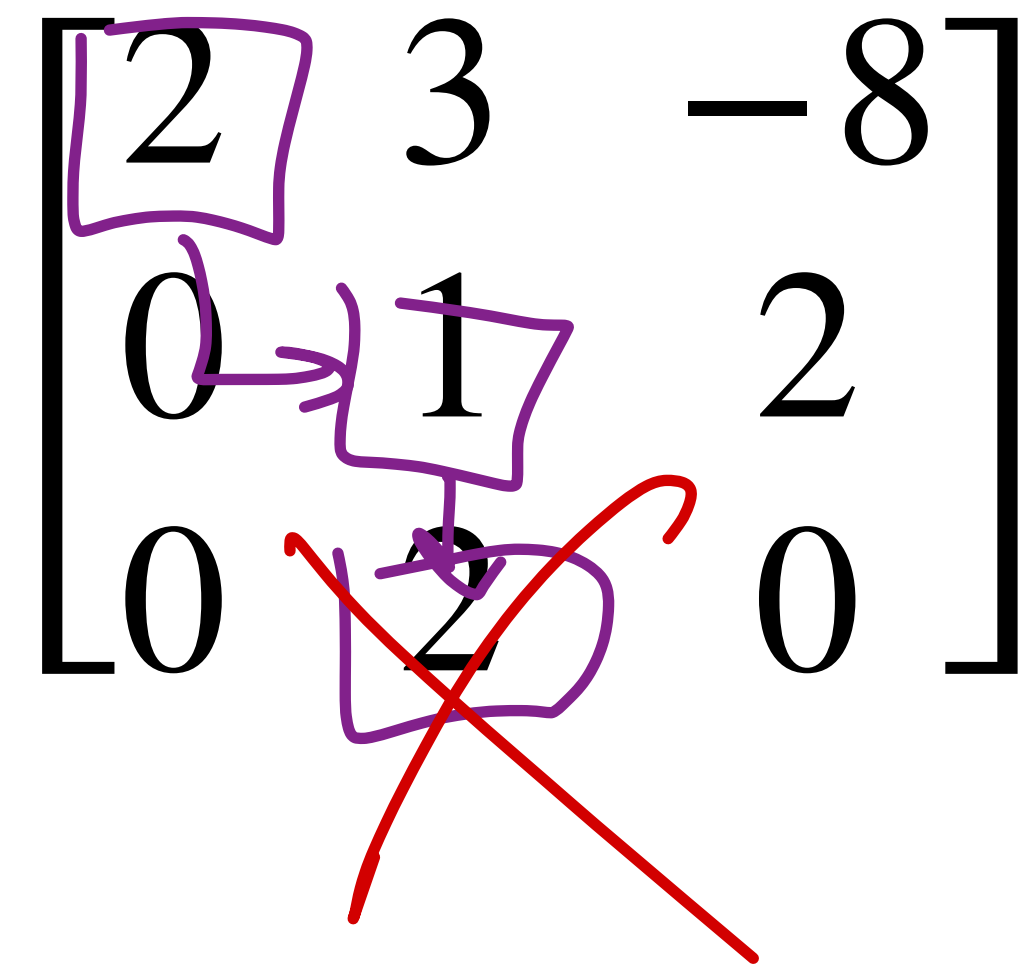
Answer: Yes

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

the leading entries of each row appears to the right of the leading entry above it

it has no all-zero rows

Question

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$


Is this matrix in echelon form?

Answer: No

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

The leading entry of the least row is not to the right of the leading entry of the second row

What's special about Echelon forms?

Thm. Let A be the augmented matrix of an *inconsistent* linear system. If $A \sim B$ and B is in echelon form then B has the row

$$[0 \ 0 \ \dots \ 0 \ 0 \ \blacksquare]$$

What's special about Echelon forms?

Thm. Let A be the augmented matrix of an *inconsistent* linear system. If $A \sim B$ and B is in echelon form then B has the row

$$[0 \ 0 \ \dots \ 0 \ 0 \ \blacksquare]$$

If all we care about is consistency then we just need to find an echelon form

Example

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ -1 & 1 & 5 & -3 \\ 1 & 2 & 4 & 7 \end{bmatrix}$$

Handwritten annotations: $+1$ above -1 , $\times 0$ above 1 , -2 above 5 , $+4$ above -3 . The -1 and 1 in the first column are circled in blue.

$$R_2 \leftarrow R_2 + R_1$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & 1 \\ 1 & 2 & 4 & 7 \end{bmatrix}$$

Handwritten annotations: -1 above 1 , -0 above 2 , $+2$ above 4 , -4 above 7 .

$$R_3 \leftarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 2 & 6 & 3 \end{bmatrix}$$

Handwritten annotations: -2 above 2 , $+6$ above 6 , -2 above 3 .

$$R_3 \leftarrow R_3 - 2R_2$$

$$\begin{aligned} x - 2z &= 4 \\ -x + y + 5z &= -3 \\ x + 2y + 4z &= 7 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Handwritten annotations: The third row is highlighted in yellow. Arrows point from the boxed 1 in the first row to the boxed 1 in the second row, and from the boxed 1 in the second row to the boxed 1 in the third row.

NO SOLUTION

The Problem with Echelon Forms

If our system *is* consistent, we can't "read off" a solution from an echelon form

Reduced Echelon Form

Row-Reduced Echelon Form (RREF)

Definition. A matrix is in *(row-)reduced echelon form* if

1. The leading entry of each row appears to the right of the leading entry above it
2. Every all-zeros row appears below any non-zero rows
3. The leading entries of non-zero rows are 1
4. the leading entries are the only non-zero entries of their columns

Reduced Echelon Form (Pictorially)

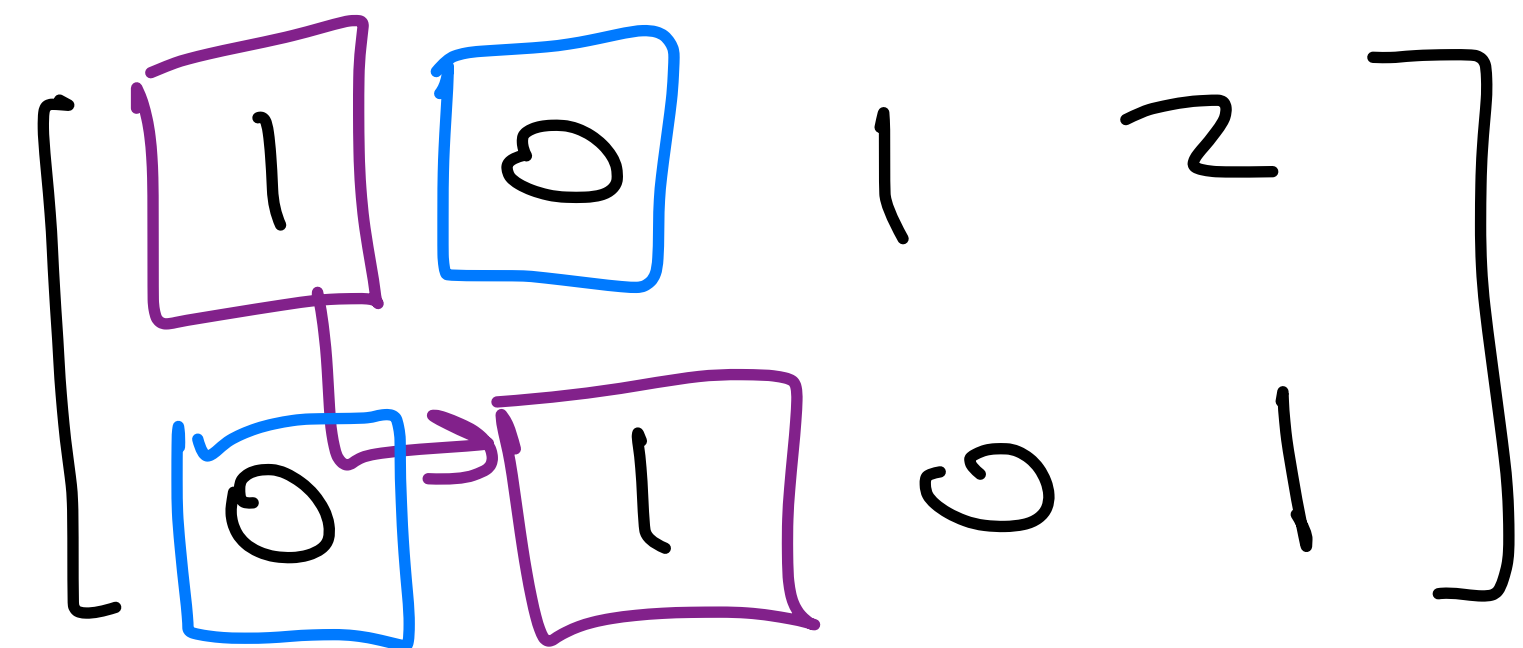
leading entries are 1

other column entries are 0

$$\begin{bmatrix} 0 & \boxed{1} & * & 0 & 0 & 0 & * & * & \boxed{0} & * \\ 0 & 0 & 0 & \boxed{1} & 0 & 0 & * & * & \boxed{0} & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & \boxed{0} & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & \boxed{0} & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \boxed{0} & 0 \end{bmatrix}$$

The diagram illustrates a matrix in reduced echelon form. The matrix is a 6x10 grid. The leading entries (1s) are highlighted in light blue boxes: the 1 in row 1, column 2; the 1 in row 2, column 4; and the 1 in row 5, column 9. Blue arrows point from the text "leading entries are 1" to these three 1s. Another set of blue arrows points from the text "other column entries are 0" to the 0s in columns 2, 4, and 9, specifically the 0s in rows 2, 3, 4, 5, and 6 of these columns. The matrix is enclosed in large square brackets on the left.

Basic Example



$$x_1 + x_3 = 2$$
$$x_2 = 1$$

Basic Example

$$x_1 + x_3 = 2$$

$$x_2 = 1$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

The Fundamental Theorem

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Thm. *every* matrix is row equivalent to a *unique* matrix in reduced echelon form

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The Point. we can "read off" the solutions of a system of linear equations from its RREF

How-To: Solving a System of Linear Equations

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1. Write your system as an augmented matrix

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1. Write your system as an augmented matrix
2. Find the RREF of that matrix

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3. Read off the solution from the RREF

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1. Write your system as an augmented matrix

2. Find the RREF of that matrix

3. Read off the solution from the RREF

Our next topic

What's special about RREF?

Every leading variable can be written in terms of only non-leading variables

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

A 6x10 matrix in row echelon form. The first row has a leading 1 in the second column. The second row has a leading 1 in the fourth column. The third row has a leading 1 in the fifth column. The fourth row has a leading 1 in the sixth column. The fifth row has a leading 1 in the ninth column. The sixth row is all zeros. Blue circles highlight the leading 1s in the second, fourth, fifth, sixth, and ninth columns. A purple oval highlights the entire second row. Two green circles highlight the asterisks in the seventh and eighth columns of the second row.

$$\begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

A 2x3 augmented matrix. The first column is labeled x_1 , the second column is labeled x_2 , and the third column is labeled x_3 . The first row has a 1 in the first column, a 0 in the second column, and a 1 in the third column. The second row has a 0 in the first column, a 1 in the second column, and a 0 in the third column. The right-hand side of the augmented matrix is 2 for the first row and 1 for the second row. Blue circles highlight the leading 1s in the first and second columns. Purple and orange ovals group the columns under x_1 , x_2 , and x_3 respectively.

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Forward Elimination and Back Substitution (Revisited)

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The goal of forward-elimination is to reduce matrix to an echelon form

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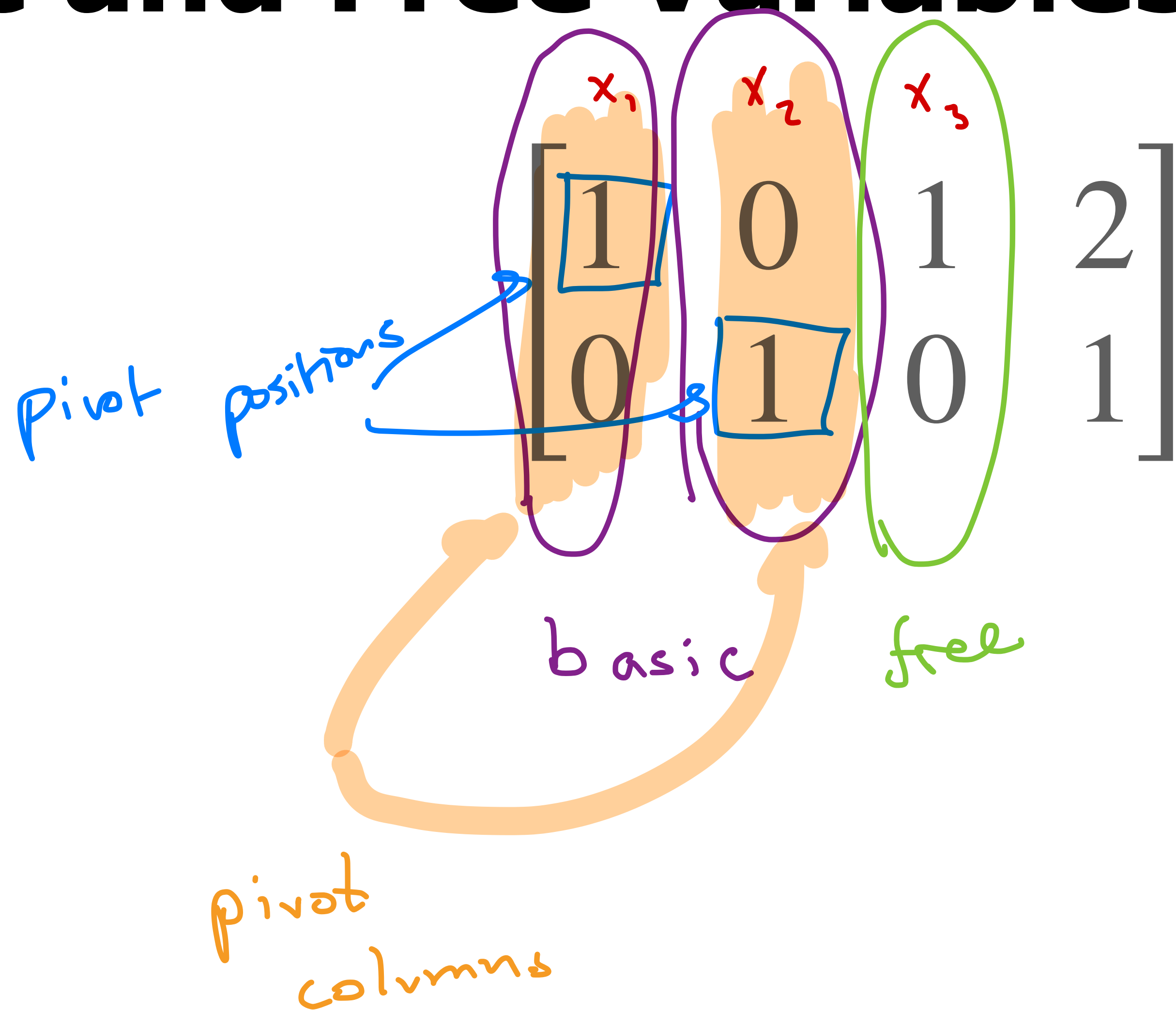
The goal of back-substitution is to reduce an echelon form matrix to its **reduced** echelon form

General-Form Solutions

Q: How do we characterize all solutions, particularly in the infinite solution case?

Basic and Free Variables

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Df. a ***pivot position*** (i,j) in a matrix is the position of a leading entry in it's reduced echelon form

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The diagram shows a matrix in reduced echelon form:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Annotations:

- x_1 is basic (points to the first column, which has a pivot at row 1)
- x_2 is basic (points to the second column, which has a pivot at row 2)
- x_3 is free (points to the third column, which does not have a pivot)

Solutions of Reduced Echelon Forms

the row i of a pivot position describes the value of x_i in a solution to the system, in terms of the free variables

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

How-To: General Form Solution

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

How-To: General Form Solution

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

1. For each pivot position (i,j) , isolate x_j in the equation in row i

How-To: General Form Solution

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$x_1 = 2 - x_3$$

$$x_2 = 1$$

x_3 is free

1. For each pivot position (i,j) , isolate x_j in the equation in row i

2. If x_i is not in a pivot column then write

x_i is free

Example

$$x_1 + 2x_2 - 2x_3 = 4$$

The diagram shows an augmented matrix with four columns and three rows. The first column is enclosed in a green oval and labeled 'basic' below it. The second and fourth columns are enclosed in red ovals and labeled 'free' below them. The third column is enclosed in a green oval. The first row has a blue box around the '1' in the first column and a blue arrow pointing to the '1' in the third column. The second row has a blue box around the '1' in the third column. The third row is highlighted with a thick orange horizontal bar. Below the matrix, the variables x_1 , x_2 , x_3 , and x_4 are written in red, corresponding to the columns.

$$\begin{bmatrix} 1 & 2 & 0 & -2 & 4 \\ 0 & 0 & 1 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

x_1 x_2 x_3 x_4

basic free

Question

$$\begin{bmatrix} 1 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Circle the pivot positions, highlight the pivot rows.

Which variables are free? Which are basic?

Write down a solution in general form for this reduced echelon form matrix.

*Write down a **particular** solution given the general form.*

Answer

$$\begin{bmatrix} 1 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 1 - 3x_4$$

x_2 is free

$$x_3 = 4 - 2x_4$$

x_4 is free

Summary

Echelon form matrices "represent solutions"

General form solutions can be used to describe the infinite solution sets